

Bumblebees like flowers...presumably?

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Data Availability: Version-controlled code necessary to reproduce the analyses can be found at <https://github.com/jmelanson98/fvpollens>. Full datasets will be shared to the same repository and archived on Dryad following acceptance.

1

Abstract

2

abstract text

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1 Introduction

For highly mobile pollinators like bumblebees, colonies acquire food resources (nectars and pollens) within a foraging kernel that can span multiple square kilometres (Melanson et al., 2026; Mola & Williams, 2025; Osborne et al., 2008). In agroecosystems, these foraging kernels encompass a variety of landcover types (e.g., managed crop fields, field or road margins, seminatural habitat fragments) which are likely to provide different floral resources at different times throughout the colony cycle. The importance of landscape-scale floral resource availability for colony growth and reproduction has been demonstrated previously (Rundlöf et al., 2022; Williams et al., 2012), as have the impacts of landscape composition and connectivity on colony-level pollen collection (CITE).

At the level of the individual forager, pollen collection is likely to arise from both local characteristics of the floral assemblage where the forager is captured, as well as landscape-scale habitat availability; importantly, the relative contributions of these two processes may inform our understanding of pollinator behaviour in agricultural landscape mosaics. For example, if foragers show high fidelity to single resource patches (as demonstrated in Woodgate et al. (2017)), local-scale patterns of floral richness and abundance will best predict the floral richness and turnover of bee-collected pollen loads. In contrast, foragers which link together multiple resource patches in the course of a single foraging trip (Lihoreau et al., 2012) may collect pollen loads that reflect landscape-scale patterns of floral resource or habitat availability.

In both cases, floral fidelity is an important secondary consideration; bumble bees have long been thought to “major and minor” on floral species, primarily visiting a highly-rewarding focal (“major”) plant species while monitoring the rewards offered by additional non-focal (“minor”) plants (Heinrich, 2004). This behaviour is intuitive from a cognitive standpoint, with multiple lines of evidence suggesting that floral con-

30 stancy should be the norm (e.g., familiar flowers may provide a stronger search image
31 and allow for more time-efficient extraction of nectar and pollen (Chittka, 2023)). This
32 expectation is supported by empirical evidence highlighting the consistency of mean
33 pollen load species richness across landscapes, timepoints, and bumble bee species
34 in UK farmlands (CITE TIMBERLAKE TENADAY). However, substantial variation in
35 pollen load richness occurred around the population mean, suggesting that additional
36 processes—such as the local or landscape-scale availability of the “major” plant—may
37 govern floral constancy.

38 Finally, bumblebees are eusocial species, which collect pollen primarily to provision
39 their young (Goulson, 2010). If pollen load species richness is behaviourally con-
40 strained at the individual level, as previous evidence suggests, individual pollen load
41 richness (“alpha diversity”) may be an uninformative measure of colony- or population-
42 level intake (“gamma diversity”), which is a proximate determinant of colony growth
43 and reproduction. To better understand the impacts of local and landscape-scale re-
44 source availability on pollinator diet breadth, we must also consider the composi-
45 tional turnover between foragers (“beta diversity”) which may be higher at florally
46 rich sites or in landscapes offering high-quality or diverse habitat types. Not only
47 will this approach provide a clearer understanding of how multiscale resource avail-
48 ability impacts pollinator diet breadth, but will also outline the behavioural processes
49 underpinning this variation, which may tradeoff across space or time.

50 2 Methods

51 2.1 Study System

52 2.2 *Bombus* and Floral Surveys

53 2.3 Pollen Metabarcoding

54 2.3.1 Pollen Collection and Quantification

55 For all *B. mixtus* and *B. impatiens* workers, we recorded the presence or absence of cor-
56 bicular pollen as a binary variable. We then removed pollen from the corbiculae of all
57 *B. mixtus* workers and a subset of *B. impatiens* workers; *B. impatiens* workers were se-
58 lected from the time period which maximized overlap with *B. mixtus* worker activity
59 (18 July-16 August) (Fig. A1). This selection of samples allowed us to draw conclu-
60 sions about season-long and interannual variation in *B. mixtus* pollen collection, while
61 also allowing us to make comparisons between the two species—which have differing
62 phenologies—during a period of temporal overlap when resource availability was the
63 same for both species.

64 We measured the mass of collected pollen loads using a microbalance (GET BRAND)
65 and stored the pollen at -80°C until DNA extraction. In cases where only a single
66 pollen load was present (e.g., because the second pollen load was dislodge during bee
67 collection) we multiplied the mass of the single pollen load by two to estimate the
68 total mass of pollen collected by the bee.

69 2.3.2 DNA Extraction, Amplification, and Sequencing

70 2.3.3 Bioinformatics

71 We processed ITS2 amplicon sequencing data using the DADA2 pipeline (Callahan,
72 n.d.; Callahan et al., 2016) implemented in R version 4.3.3 (R Core Team, 2021) with li-
73 braries *dada2* (Callahan et al., 2016), *Biostrings* (Pagès et al., 2026), and *ShortRead* (Mor-
74 gan et al., 2009). Given the variable length of ITS sequences across species, we filtered

75 the demultiplexed reads using $\text{maxN} = 0$ but no sequence truncation. We used cu-
76 tadapt (Martin, 2011) to trim primer sequences, then performed a second round of
77 quality filtering using the parameters $\text{minLen} = 50$, $\text{maxN} = 0$, $\text{maxEE} = c(2,2)$, $\text{truncQ} =$
78 2 (DADA2 default parameters (Callahan, n.d.)). The DADA2 pipeline estimates the se-
79 quence error rates to probabilistically discriminate sequences real sequences from those
80 arising through sequencing errors; we used the `learnErrors` function to generate a
81 parametric model of sequence errors, which was used to perform inference of the
82 assigned sequence variants (ASVs) using the `dada` function (Callahan et al., 2016).
83 Finally, we removed chimeric sequences using the `removeBimeraDenovo` function
84 (Callahan et al., 2016); for all three sequencing runs, the percentage of reads identi-
85 fied as chimeric was between 9-11%

86 The output of the DADA2 pipeline is a table of ASVs (i.e., unique sequences) and their
87 abundances across samples. We assigned taxonomy to ASVs in QIIME2 (Bolyen et al.,
88 2019)

89 2.4 Statistical Analyses

90 Some ideas for analysis:

- 91 • colony-level bumblebee-floral networks (see (Ye et al., 2024)) – but find some
92 way to account for phenology

93 3 Results

94 4 Discussion

95 References

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156 **Supporting Information**